



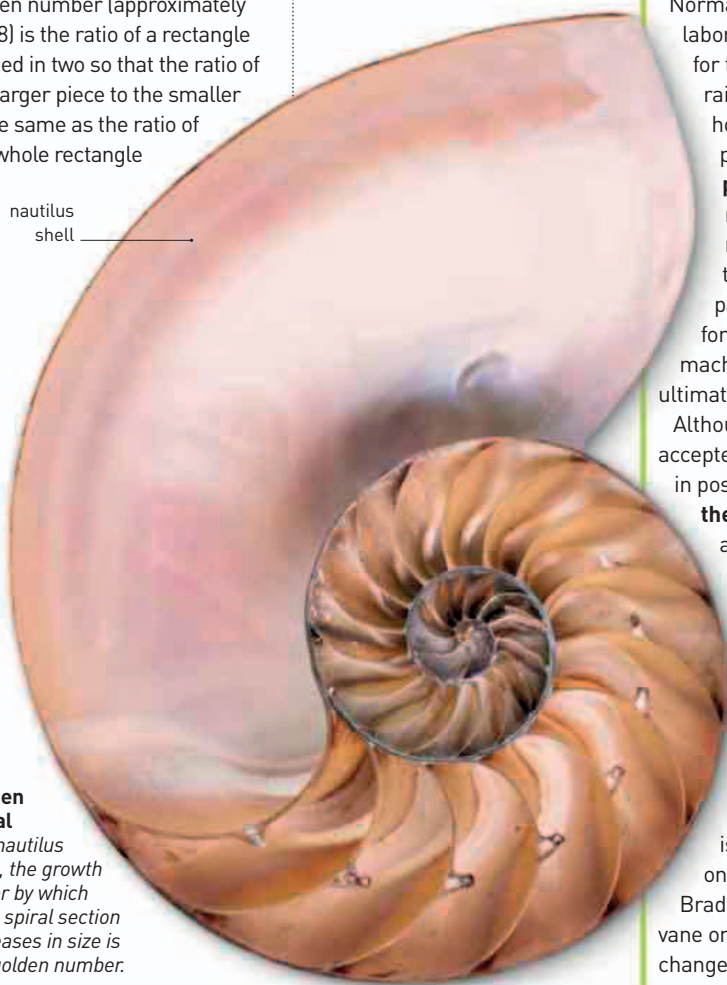
William Ged's stereotype printing process involved making a copy of the typeset page from a mold, so the copy could be used again and again for reprints.

(1700–82) as professor. That year, Bernoulli linked two ancient concepts: the **golden number**, which the Ancient Greeks believed gave perfect artistic proportions, and the **Fibonacci sequence** (see 1200–19). The golden number (approximately 1.618) is the ratio of a rectangle divided in two so that the ratio of the larger piece to the smaller is the same as the ratio of the whole rectangle

to the larger. In the Fibonacci sequence, each number is the sum of the previous two numbers. Bernoulli showed that the golden number is in fact the ratio of any Fibonacci number to the previous number in the sequence.

nautilus shell

Golden spiral
In a nautilus shell, the growth factor by which each spiral section increases in size is the golden number.



FROM THE 16TH CENTURY, Lyon in France had been the center of European **silkmaking**. It was here in 1725 that silkmaker **Basile Bouchon** invented a system for setting up the cords on the silk loom.

Normally, this was a long and laborious job, but by arranging for threading needles to be raised or not according to holes on a moving roll of paper, Bouchon could **partially automate** the machine. This reduced mistakes, speeding up the process. Bouchon's paper roll paved the way for all programmable machines, including, ultimately, today's computers.

Although by 1700 it was widely accepted that **Earth** is not fixed in position but **moves around the Sun**, it was hard to actually prove. Then, in 1725, English astronomer **James Bradley** (1693–1762) observed the star Gamma Draconis moving in the opposite direction to the way it usually did. This was difficult to explain, but it is said that while sailing on the Thames River

Bradley realized the weather vane on the mast sometimes changed direction not because

10 THOUSAND THE NUMBER OF VOLUMES IN THE GUJIN TUSHU JICHENG

the wind's direction was changing but because the boat was changing its course. In the same way, Bradley surmized, the mysterious change in the direction of the stars, now known as **stellar aberration**, must be caused by the changing motion of Earth. In London this year, Scottish printer **William Ged** (1699–1749) invented the **stereotype**—a copy of an original typeset page made using a mold. This meant that limitless copies could be made from the stereotype without the trouble of laboriously resetting the type.

Meanwhile in China, the *Gujin Tushu Jicheng* (Collection of Pictures and Writings) was being printed. It was a vast encyclopedia overseen by the Qing Dynasty emperors Kangxi and Yongzheng. Only 64 copies were ever printed, but it consisted of 10,000 volumes,

Blood pressure measurement
English clergyman Stephen Hales inserted an 11½-ft- (3.5-m-) long glass tube into the neck artery of a horse, and held it vertically to see how far the blood rose up the tube.

800,000 pages, and 100 million Chinese characters.

In 1726, English clergyman Stephen Hales (1677–1761) described how he made the **first measurements of blood pressure** by observing how far blood rose up a tube inserted in the artery of a horse. He measured the **heart's capacity and output** in various animals, and the **speed and resistance of blood flow** in the arteries.



1723 French naturalist **Antoine de Jussieu** shows that ceratonia stones are ancient tools

1724 St. Petersburg Academy of Sciences is founded

1724 Swiss mathematician **Daniel Bernoulli** links the golden number and Fibonacci sequence

1725 British astronomer **James Bradley** observes stellar aberration

1725 French silkmaker **Basile Bouchon** builds the first semiautomated machine

1725 The *Gujin Tushu Jicheng* (Collection of Pictures and Writings) is printed in China

1725 William Ged invents stereotype printing

1726 English clergyman **Stephen Hales** describes the first blood pressure measurements